



Multi-Cloud data platform integration with OneLake Shortcut and Caching

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Sponsors & Organizers



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+15 years in Microsoft “Data Realm”

Giving my little contribution to the community



DATA SATURDAYS

Dev>Marche

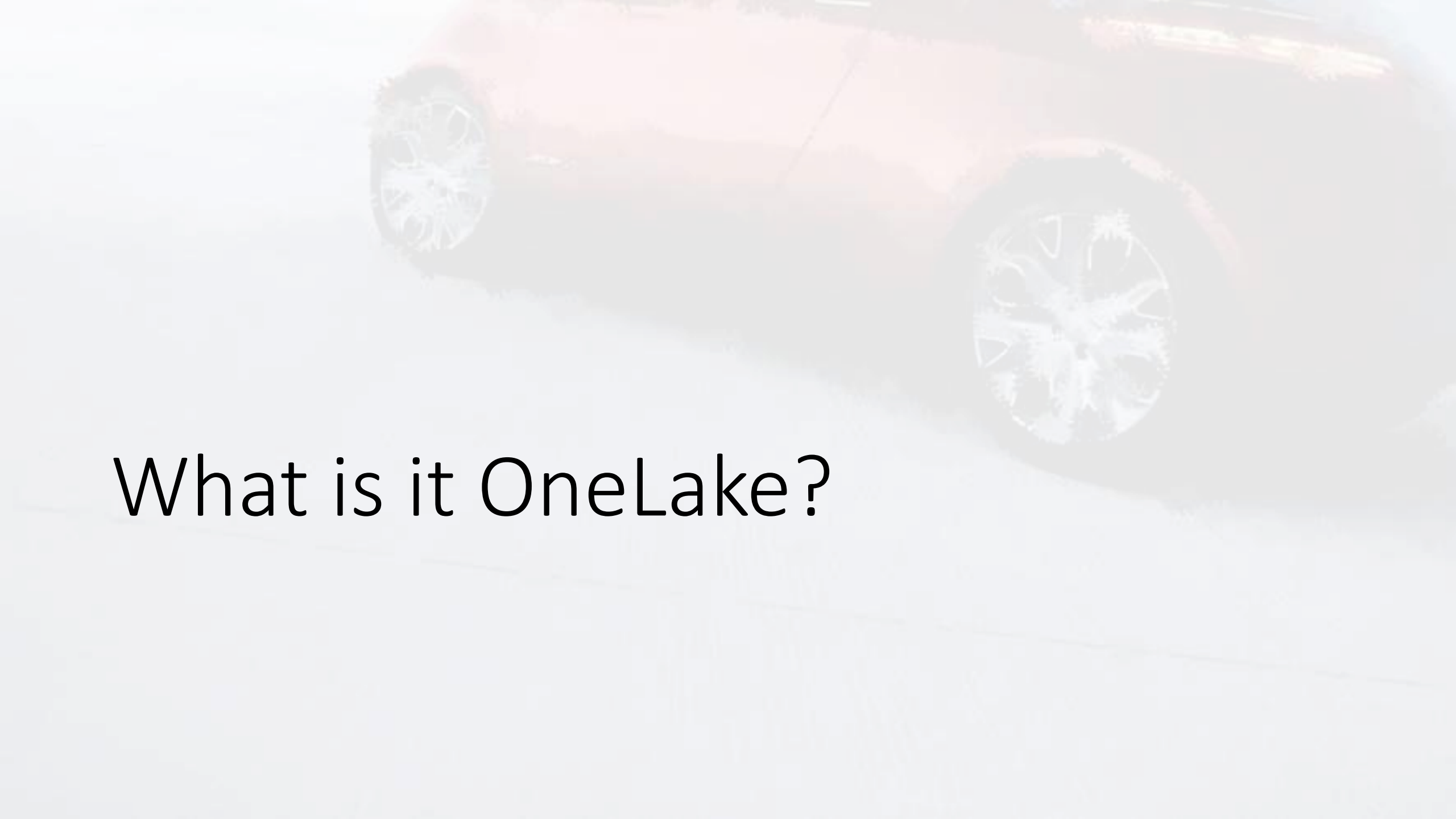
“OneLake is the
least “shiny”
Microsoft Fabric
component but
it’s most the
precious”

Riccardo Perico 😊



1 oz Gold = 4300 USD
1 oz Rhodium = 8800 USD
(Aug '26)

1 oz Californium = 800 Million USD



What is it OneLake?

OneLake for all data

“The OneDrive for data”



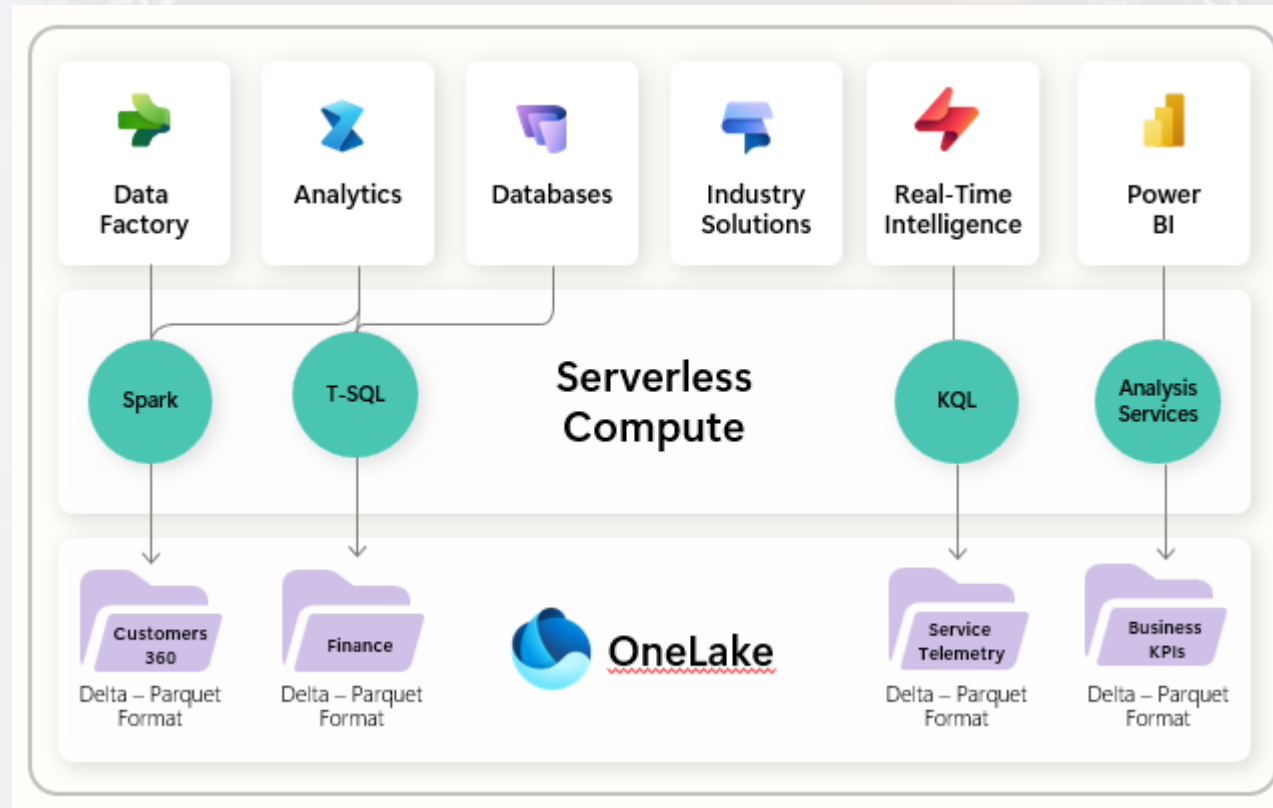
OneDrive



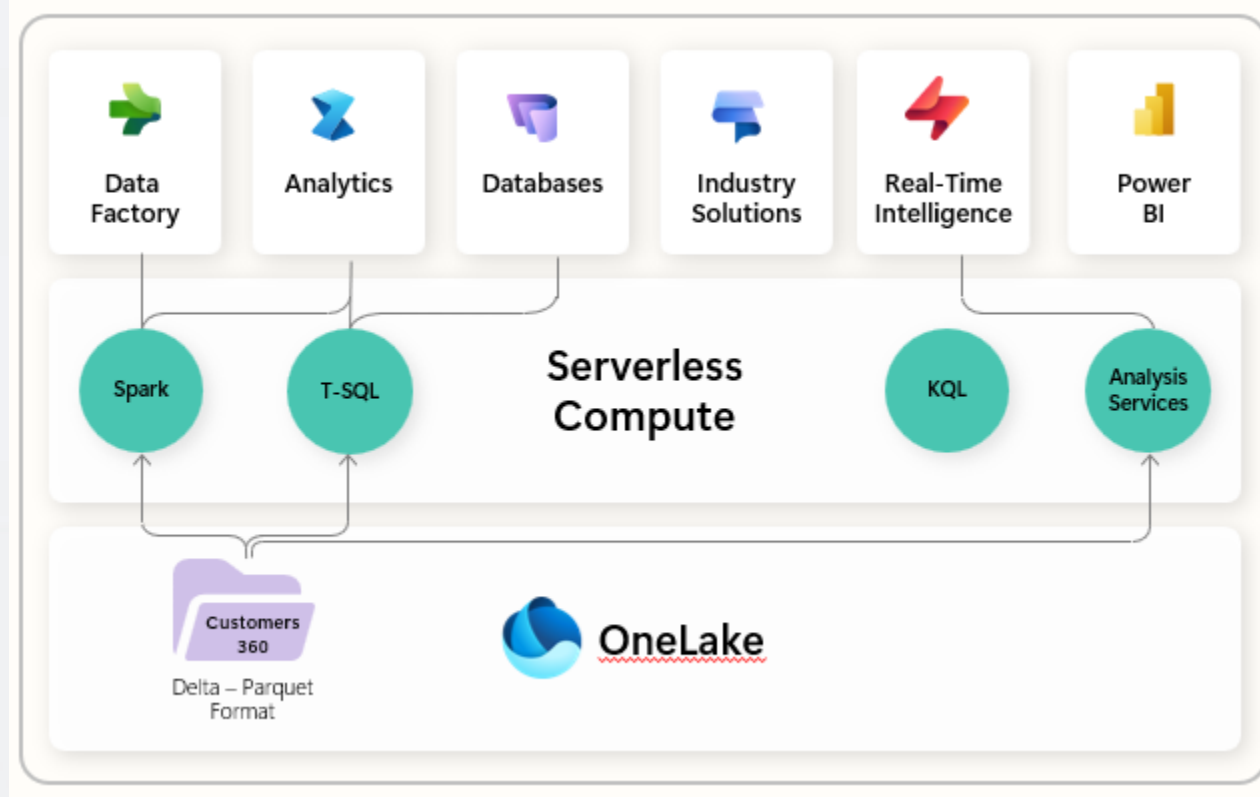
OneLake

OneLake provides a data lake as a service without you needing to build it

All Fabric Workloads can write in it

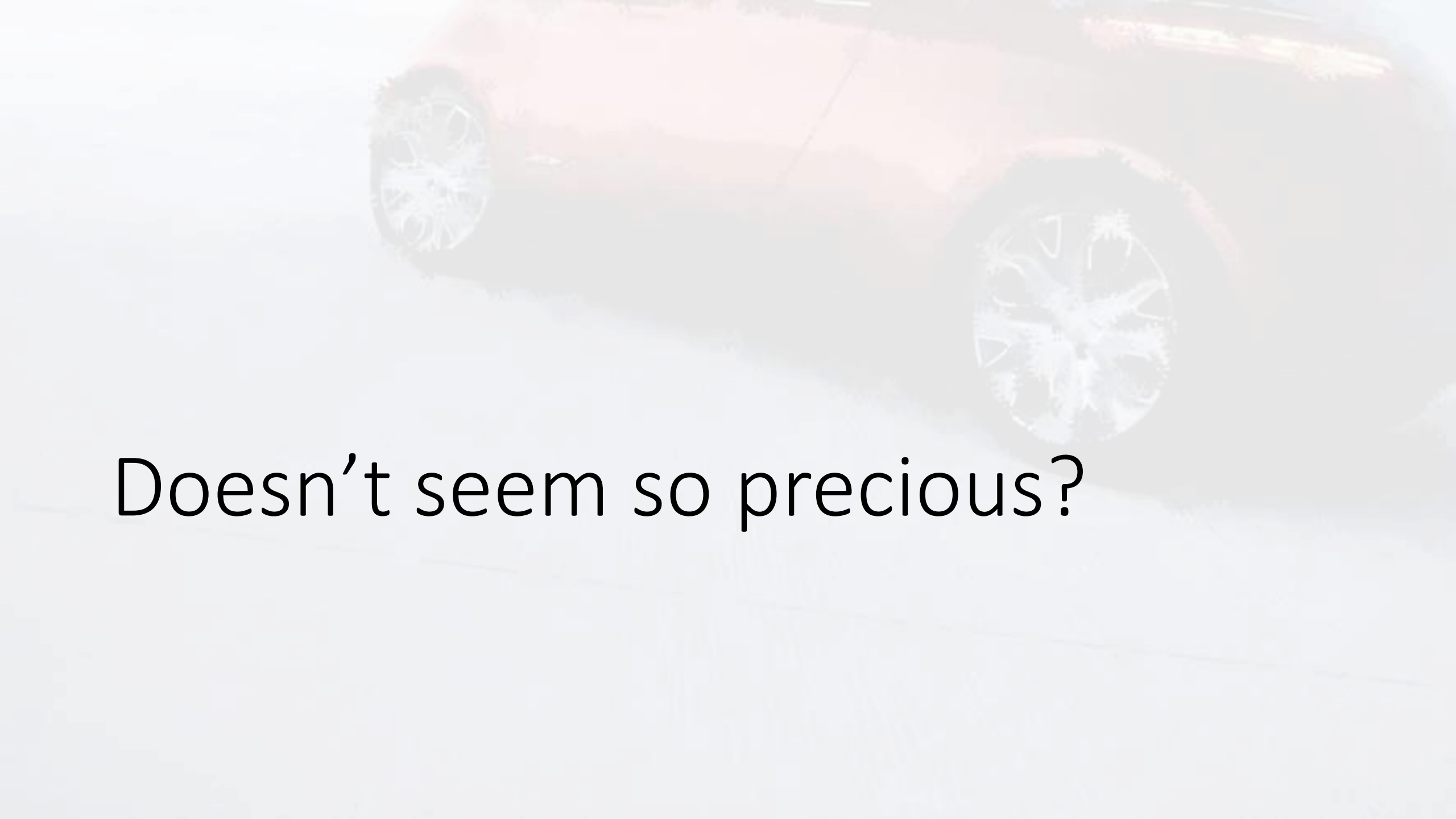


All Fabric Workloads can read from it



Revolutionary decision by Microsoft





Doesn't seem so precious?

NOW (TODAY)

NEW BUILDING, HIDDEN RISK



BEAUTIFUL
NEW
SKYSCRAPER

**BAD GROUND
EVEN TODAY!**



VERY WEAK SOIL



HIGH WATER
TABLE



LOOSE &
COMPRESSIBLE
SOIL



POSSIBLE VOIDS
& SETTLEMENT



LOOKS PERFECT TODAY...
BUT THE GROUND IS **UNSTABLE!**



THEN (FUTURE)

TIME PASSES, PROBLEMS APPEAR



BUILDING
STARTS TO
TILT



CRACKS
DEVELOP



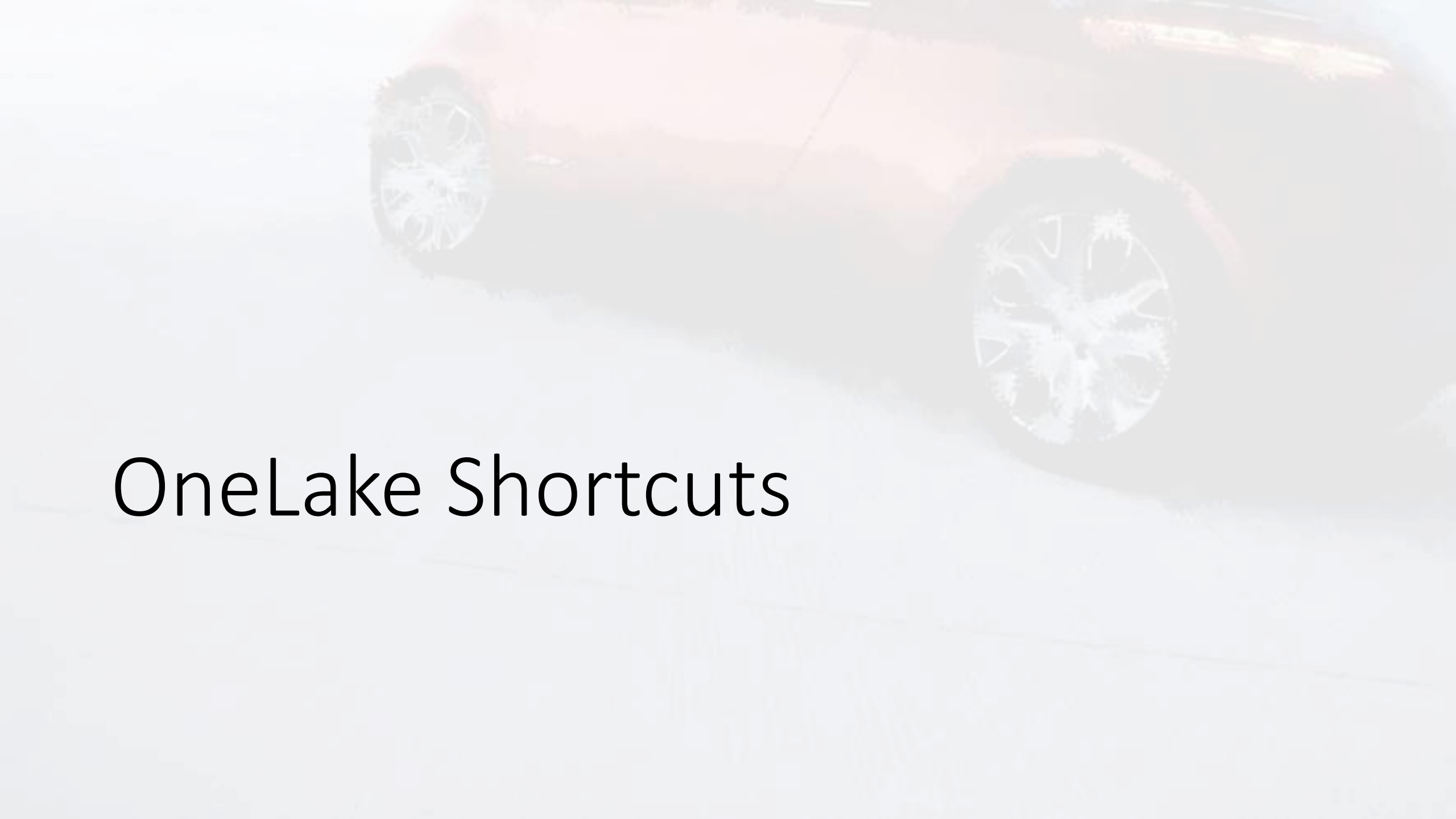
DOORS &
WINDOWS
DON'T CLOSE
PROPERLY



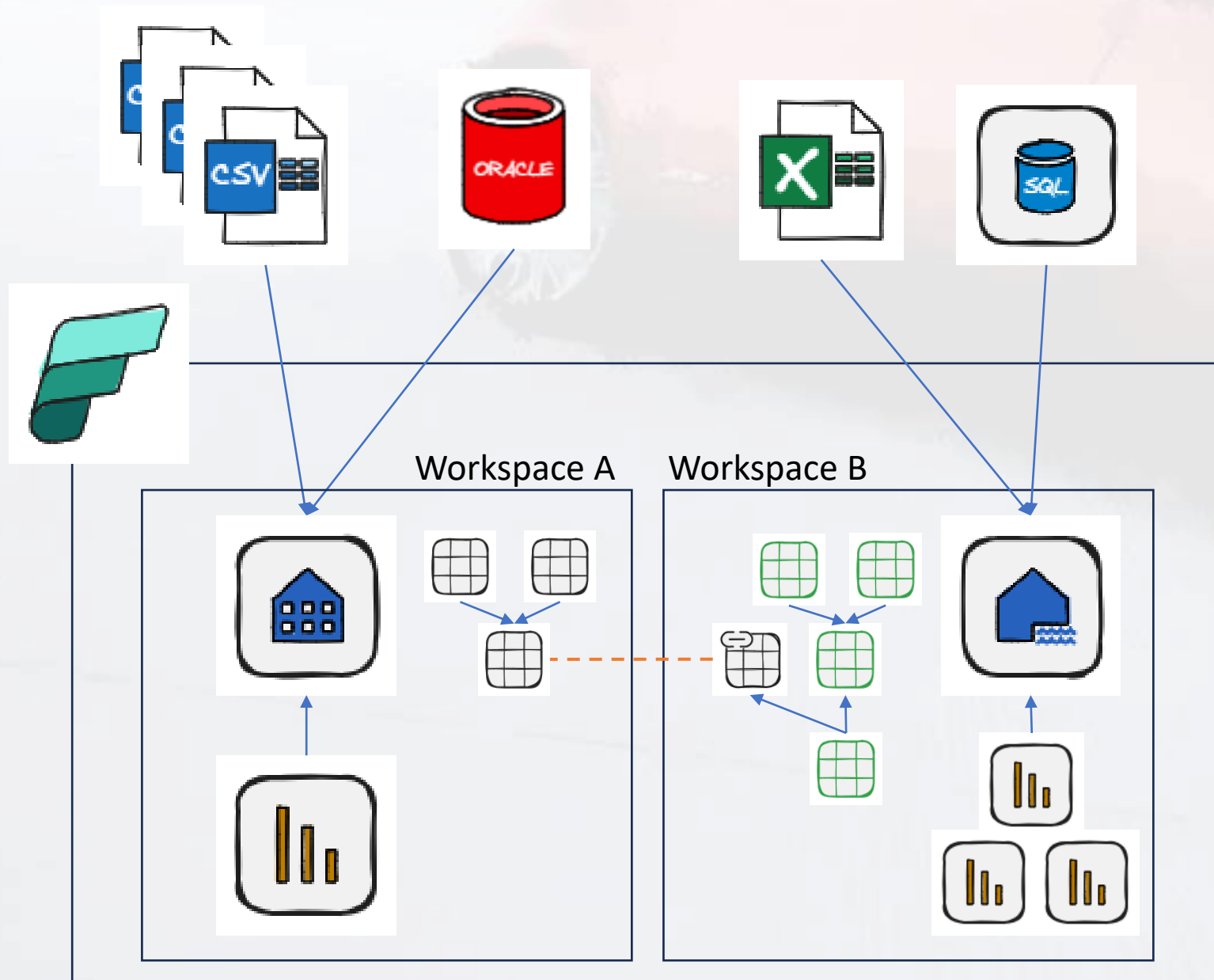
UNEQUAL
SETTLEMENT
DAMAGES
THE STRUCTURE



OVER TIME, THE BAD GROUND
CAUSES **SERIOUS PROBLEMS**
AND **HIGH COSTS!**



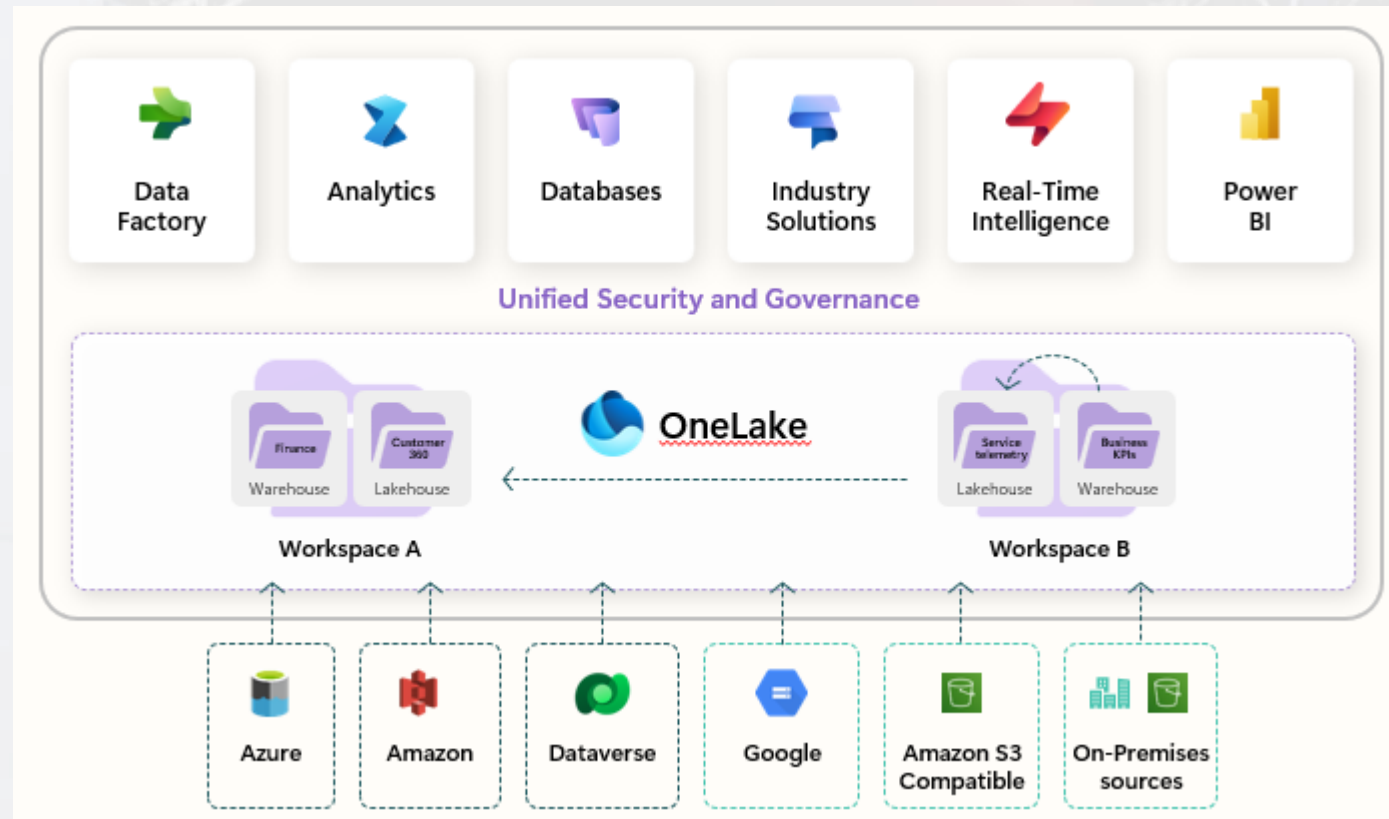
OneLake Shortcuts



Benefits:

1. Space saving
2. Time saving
3. CUs saving
4. Don't need to reinvent the wheel
5. Data ownership delegated
6. Fix once and for all
7. Add feature once and for all

Let's make the concept bigger



Networking is a factor

- Fabric to Shortcut source path could be long (Italy North -> West US) and **performance** may be impacted
- **Cloud providers charge you** for internet egress

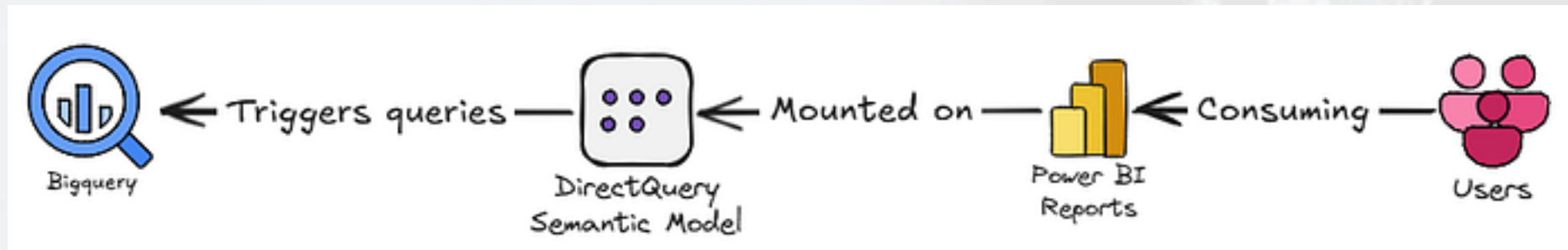
Example on 1 Shortcut reading from AWS*

	Day 1				Day 2				Day 3	
	Query 1 10 GB	Query 2 30 GB	Query 4 50 GB	Query 5 1 GB	Query 1 10 GB	Query 2 30 GB	Query 4 50 GB	Query 5 1 GB		
		Query 3 10 GB				Query 3 10 GB				
Data Read	10 GB	50 GB	100 GB	101 GB	111 GB	151 GB	201 GB	202 GB		
Query Cost	0.00 USD	0.00 USD	0.00 USD	0.09 USD	0.90 USD	3.60 USD	4.50 USD	0.09 USD		
Total Bill	0.00 USD	0.00 USD	0.00 USD	0.09 USD	0.99 USD	4.59 USD	9.09 USD	9.18 USD		

Day 2 costed 100X Day 1

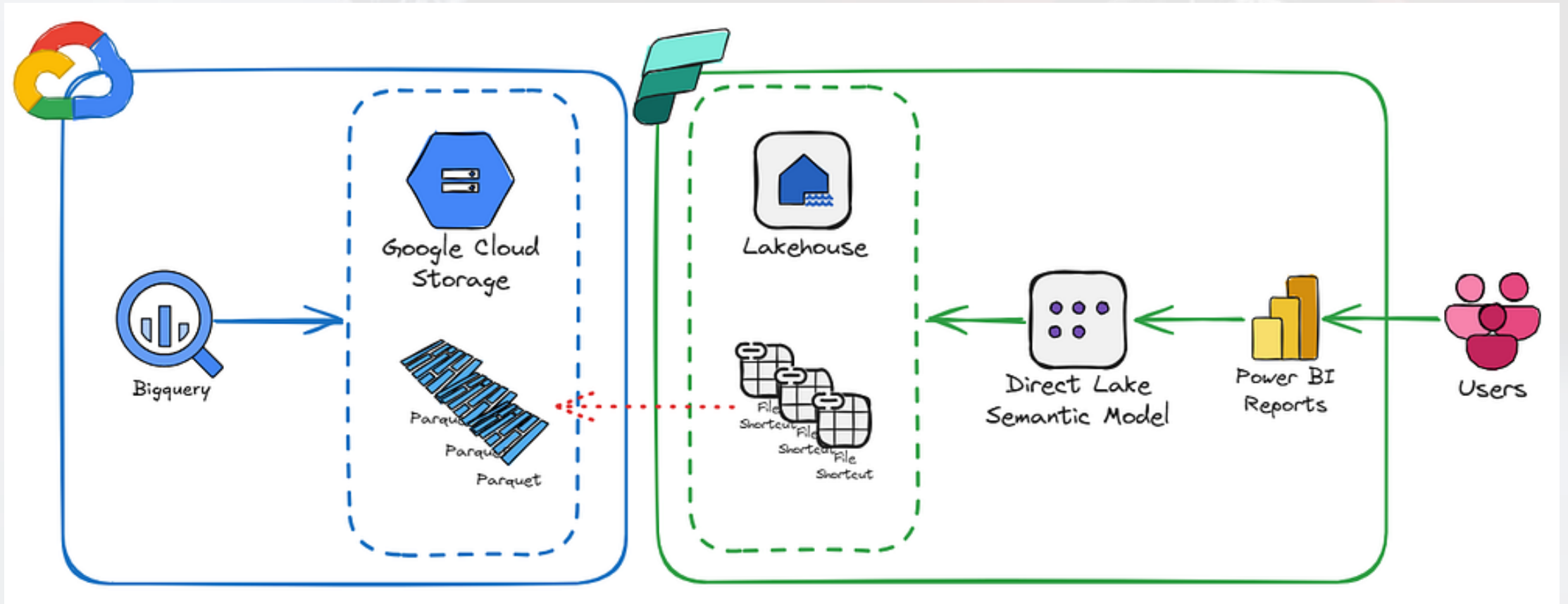
*prices may change

Scenario from the real world



Big query DWH is loaded
once per week

We proposed a rebuild



Problem: We're hitting GCS for any query of the user

OneLake caching to reduce Egress

Workspace settings

Search

- General
- License info
- Azure connections
- System storage
- Git integration
- OneLake**
- Workspace identity
- Network security
- Power BI
- Delegated Settings
- Extension API playground
- Data Engineering/Science
- Data Factory

OneLake Settings

Configure and manage settings for OneLake in this workspace

[Learn more about OneLake](#)

OneLake File Explorer

The OneLake file explorer application seamlessly integrates OneLake with Windows File Explorer

[Download OneLake app](#)

Shortcut Settings

Enable cache for shortcuts ☒ On

Data accessed through shortcuts in this workspace will be cached in OneLake. This data will remain in cache up to the defined retention period.

[Learn more about shortcuts cache settings](#)

Retention Period

Select the maximum time that data from a shortcut can be retained in cache within the workspace. You can set retention period from 1 - 28 days.

28 days

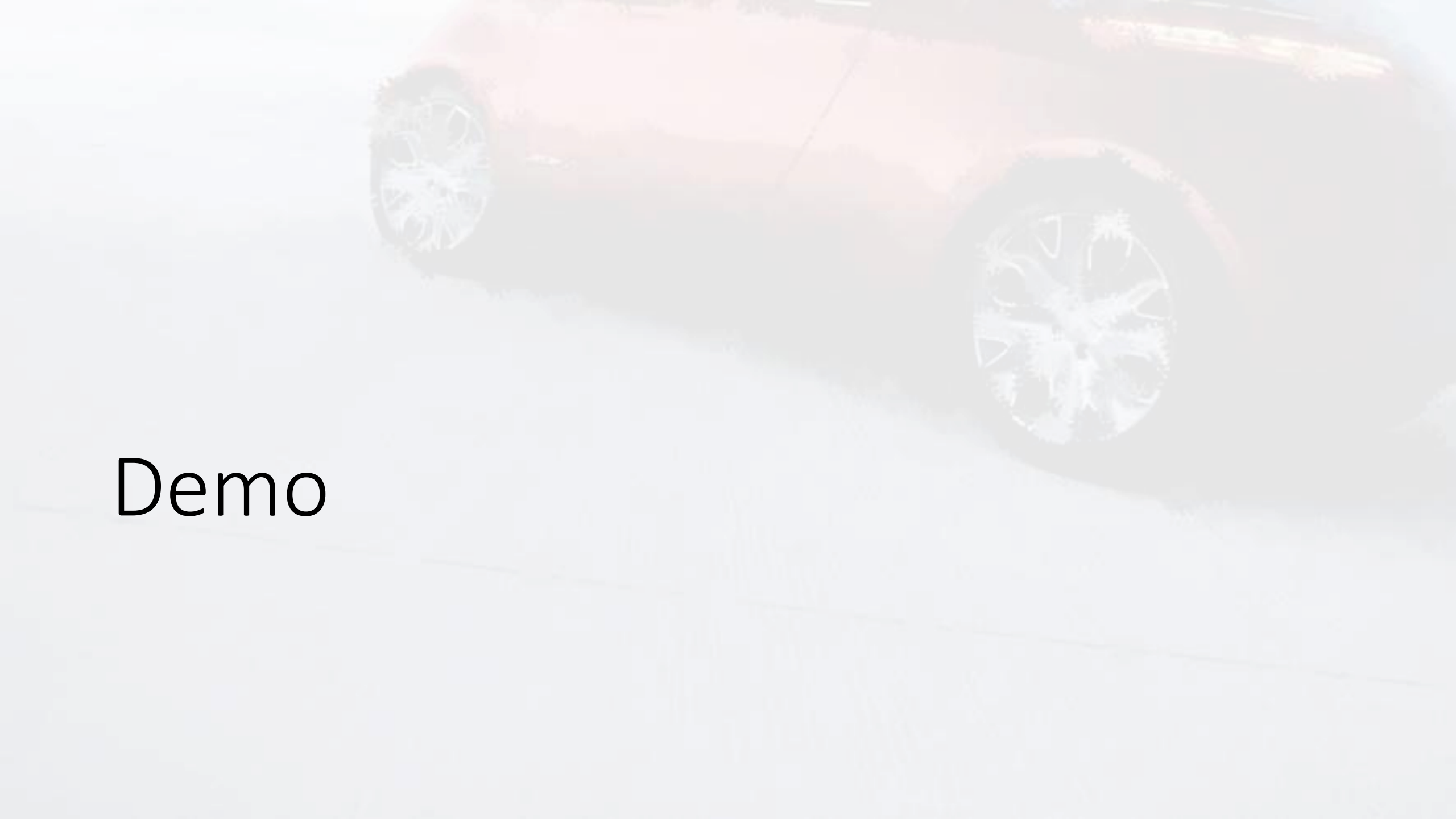
Reset cache

This action will remove all cached data for shortcuts in this workspace from OneLake.

[Reset cache](#)

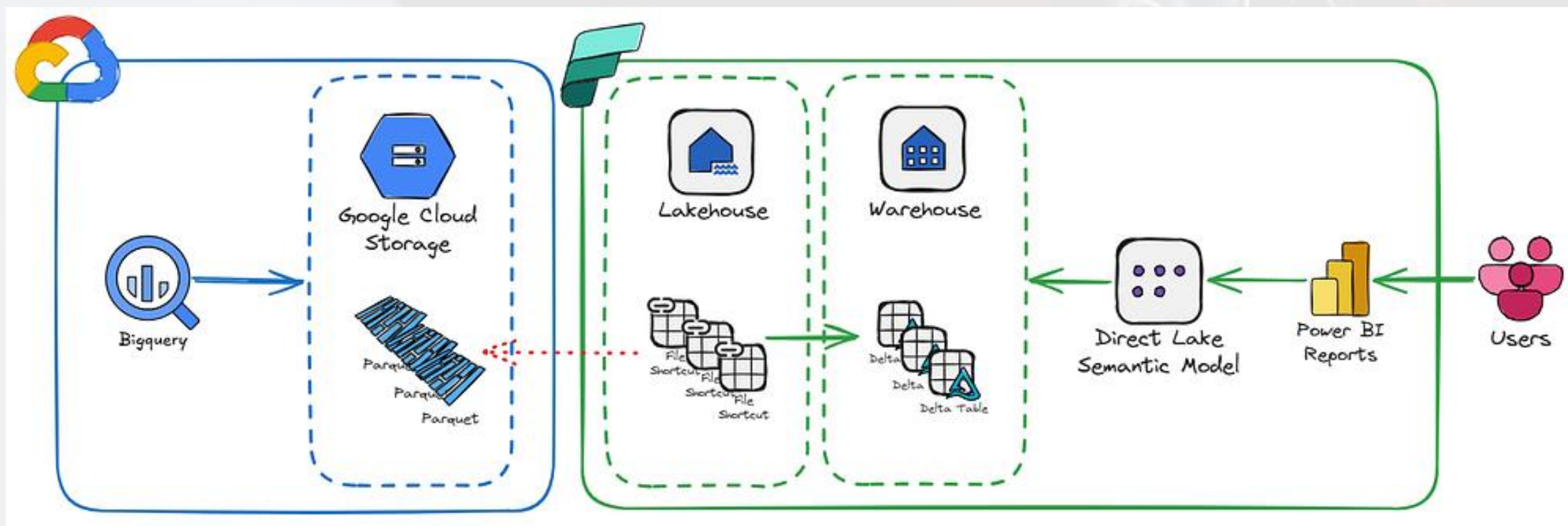
Behind the scenes

- You set the retention between 1-28 days
- Supported:
 - GCS
 - S3
 - S3 Compatible (even on-prem)
- Up to 1GB/file
- First read from source subsequent from Fabric cache unless
 - Retention reached
 - Source has more recent file



Demo

The final layout



Savings on GC: 92%

Caveats

- Cache observability → you don't know if a query hit the cache
- Cache visibility → you don't know if the cache is ready
- Cache scope → at the workspace level

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Q&A





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Thanks!